

**Multiple Choice:**

1. In a single sideband and CW receiver, the antenna is connected to the _____.
 - A. product detector
 - B. high frequency oscillator
 - C. intermediate frequency amplifier
 - D. radio frequency amplifier
2. The three main parameters against which the quality of a receiver is measured are:
 - A. selectivity, stability and frequency range
 - B. sensitivity, stability and crossmodulation
 - C. sensitivity, selectivity and image rejection
 - D. sensitivity, selectivity and stability
3. Why is FM voice best for local VHF/UHF radio communications?
 - A. It has high-fidelity audio which can be understood even when the signal is somewhat weak
 - B. The carrier is not detectable
 - C. It is more resistant to distortion caused by reflected signals
 - D. Its RF carrier stays on frequency better than the AM modes
4. Which of the following modes of transmission is usually detected with a product detector?
 - A. Double sideband full carrier
 - B. Frequency modulation
 - C. Pulse modulation
 - D. Single sideband suppressed carrier
5. SSB transmitters require a(n) _____ wattmeter and AM or FM transmitters require a(n) _____ wattmeter.
 - A. RMS-reading, peak-reading.
 - B. Peak-reading, rms-reading.
 - C. Peak-reading, PEP.
 - D. PEP, Peak-reading.
6. Most AM broadcasts employ _____ polarization while most FM broadcasts employ _____ polarization of the radio wave.
 - A. vertical, vertical
 - B. vertical, horizontal
 - C. horizontal, vertical
 - D. horizontal, horizontal
7. In a frequency modulation transmitter, the input to the speech amplifier is connected to the:
 - A. Microphone
 - B. modulator
 - C. power amplifier
 - D. frequency multiplier
8. How can intermodulation interference between two transmitters in close proximity often be reduced or eliminated?
 - A. By using a Class C final amplifier with high driving power.
 - B. By installing a terminated circulator or ferrite isolator in the feed line to the transmitter and duplexer.
 - C. By installing a band-pass filter in the antenna feed line.
 - D. By installing a low-pass filter in the antenna feed line.
9. In a single sideband transmitter, the _____ is in between the mixer and the antenna.
 - A. variable frequency oscillator
 - B. linear amplifier
 - C. balanced modulator
 - D. radio frequency oscillator
10. Which of the following is only required for CW or SSB demodulation?
 - A. BFO
 - B. BPF
 - C. VFO
 - D. VOX
11. Which emission modes suffer the most from selective fading?
 - A. CW and SSB.
 - B. FM and double sideband AM.
 - C. SSB and image.
 - D. SSTV and CW.
12. What is the name of the condition that occurs when the signals of two transmitters in close proximity mix together in one or both of their final amplifiers, and unwanted signals at the sum and difference frequencies of the original transmissions are generated?
 - A. Amplifier desensitization.
 - B. Neutralization.
 - C. Adjacent channel interference.
 - D. Intermodulation interference.
13. What is the term used to refer to the condition where the signals from a very strong station are superimposed on other signals being received?
 - A. Intermodulation distortion.
 - B. Cross-modulation interference.
 - C. Receiver quieting.
 - D. Capture effect.
14. What is the capture effect?
 - A. All signals on a frequency are demodulated by an FM receiver.
 - B. All signals on a frequency are demodulated by an AM receiver.
 - C. The strongest signal received is the only demodulated signal.
 - D. The weakest signal received is the only demodulated signal.
15. Motorboating (low frequency oscillations) in an amplifier can be stopped by:
 - A. Grounding the screen grid.
 - B. Bypassing the screen grid resistor with a 0.1 μ F capacitor.
 - C. Connect a capacitor between the B+ lead and ground.
 - D. Grounding the plate.
16. What kind of filter would you use to attenuate an interfering carrier signal while receiving an SSB transmission?
 - A. A notch filter



- B. A band pass filter
C. An all pass filter
D. A pi-network filter
17. In a typical CW transmitter, the _____ is the primary source of direct current.
- A. driver/buffer
B. power supply
C. power amplifier
D. master oscillator
18. Which of the following FM detectors does not require IF limiter stage to precede it?
- A. Foster-Seeley
B. Ratio
C. Quadrature
D. Stagger-tuned
19. What is meant by the term modulation index?
- A. The processor index.
B. The ratio between the deviation of a frequency modulated signal and the modulating frequency.
C. The FM signal-to-noise ratio.
D. The ratio of the maximum carrier frequency deviation to the highest audio modulating frequency.
20. What is the term used to refer to a reduction in receiver sensitivity caused by unwanted high-level adjacent channel signals?
- A. Intermodulation distortion.
B. Quieting.
C. Desensitizing.
D. Overloading.
21. To produce a single-sideband suppressed carrier transmission it is necessary to _____ the carrier and to _____ the unwanted sideband.
- A. filter, filter.
B. filter, cancel.
C. cancel, filter.
D. cancel, cancel.
22. What type of emission is produced when an amplitude modulated transmitter is modulated by a television signal?
- A. F3F
B. A3F
C. A3C
D. F3C
23. In a frequency modulation receiver, the _____ is in between the antenna and the mixer.
- A. audio frequency amplifier
B. high frequency oscillator
C. intermediate frequency amplifier
D. radio frequency amplifier
24. What is meant by the term deviation ratio?
- A. The ratio of the audio modulating frequency to the center carrier frequency.
B. The ratio of the maximum carrier frequency deviation to the highest audio modulating frequency.
C. The ratio of the carrier center frequency to the audio modulating frequency.
D. The ratio of the highest audio modulating frequency to the average audio modulating frequency.
25. How does the modulation index of a phase-modulated emission vary with the modulated frequency?
- A. The modulation index increases as the RF carrier frequency (the modulated frequency) increases.
B. The modulation index decreases as the RF carrier frequency (the modulated frequency) increases.
C. The modulation index varies with the square root of the RF carrier frequency (the modulated frequency).
D. The modulation index does not depend on the RF carrier frequency (the modulated frequency).
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